

flint – Firmware Burning Tool

The flint (Flash interface) utility performs the following functions:

- Burns a binary firmware image to the Flash device attached to an adapter or a switch device.
- Burns an Expansion ROM image to the Flash device attached to adapters.
- Queries for firmware attributes (version, GUIDs, UUIDs, MACs, PSID, etc.)
- Enables executing various operations on the Flash memory from the command line (for debug/production).
- Disables/enables the access to the device’s hardware registers, and changes the key used for enabling. This feature is functional only if the burnt firmware supports it.

flint Synopsis

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1 flint [OPTIONS] <command> [parameters...]
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witches Options



--allow_rom_change	Allows burning/removing ROM to/from Firmware image product version present.
-banks <banks>	Set the number of attached flash banks (banks)
-blank_guids	Burn the image with blank GUIDs and MACs (where applicable). These values can be updated later using the command (see details below)

	Commands burn
-clear_semaphore	Force clear of semaphore device. No command is allowed without flag is used. NOTE: May impact system installed flash corrupt device or an application currently using flash. Exercise caution.
--cpu_util <CPU_UTIL>	Use this flag to reduce CPU while burning. Windows or values are from (lowest CPU (highest CPU
-d[evice] <device>	Device flash connected to Commands all
-dual_image	Make the boot process burn images on flash (previously called algorithm). Command default fails process burn image (in alternative locations). Commands burn
-flash_params <type,log2size,num_of_flashes>	Use the given parameters the flash instructions reading the the flash.

	Supported parameters - Type: of the such as M25P M25P SST25 W25C W25X AT25L S25FL - log2s log2 c flash - num_ the nu the fla conne the de
--flashed_version	When speci flashed fw v fetched Commands query
-guid <GUID>	GUID base v GUIDs are automatical assigned to following va guid -> node guid+1 -> po guid+2 -> po guid+3 -> sy image GUID NOTE: port2 be assigned a single por - The HCA ig this value. Commands burn, sg

-guids <GUIDs...>	4 GUIDs must be specified here. The GUIDs specified GUIDs are assigned to the following files respectively: port1, port2, system image. NOTE: port2 must be specified even for a system image. HCA. The HCA takes this value. It is set to 0x0. Commands: burn, sg
-h[elp]	Prints this help and exits
-hh	Prints extended command help
--hmac_key <hmac_key>	Path to the file containing the HMAC key (image only).
--hsm	Flag to use HSM command. V HSM HW for encryption
-i[image] <image>	Binary image Commands: burn, verify
--ignore_dev_data	Do not attempt to take device sections from (sections with from the image only). Commands: burn

--key_uuid <uuid_file>	UUID match given private key used by the command
--key_uuid2 <uuid_file>	UUID match given private key used by the command
-log <log_file>	Prints the boot status to the specified log file
--low_cpu	When specified, CPU usage will be limited. Run time may be increased. Commands: burn, query
-mac <MAC> 1	MAC address value. 2 MACs must be specified. 2 MACs will be automatically assigned to the following values: mac -> port1 mac+1 -> port2 Commands: burn, sg
-macs <MACs...> 1	2 MACs must be specified here. 2 MACs will be assigned to port1, port2, respectively. Commands: burn, sg NOTE: -mac and -macs flags are applicable only for NVMe Ethernet ports
-no	Non-interactive mode - assume answers to all prompts

	to all questi Commands all
-no_flash_verify	Do not verif write on the
--no_fw_ctrl	Do not atten work with tl firmware Ct commands.
-nofs	Burns image failsafe mar
--openssl_engine <engine name>	Name of the engine to be the sign/rsa commands work with tl HSM hardw OpenSSL Af
--openssl_key_id <key>	Key identific string to be the sign/rsa commands work with tl HSM hardw OpenSSL Af
--output_file <string>	Output file i exporting th key from PE
-override_cache_replacement ²	Allow acces flash even if replacemen enabled. NOTE: This t often referr -ocr NOTE: This t intended fo users only. F this mode n the firmwar

<code>--private_key <key_file></code>	Path to PEM formatted private key to be used by command
<code>--private_key_label <string></code>	Flag to use with sign/import commands.
<code>--private_key2 <key_file></code>	Path to PEM formatted private key to be used by command
<code>-qq</code>	Run a quick query. When specified, the device will not perform image integrity checks during query operation, which may shorten execution time when running over slow interfaces (e.g. I2C, MTUSB). Command affected: query
<code>-s[ilent]</code>	Do not print burn progress to the console. Command affected: burn
<code>-striped_image</code>	Use this flag to indicate that the given image is in a "striped image" format. Commands affected: query verify
<code>-uid <UID></code>	5th Generation (Apple A15 Bionic and later) devices only. Set the base UID. Global MACs are derived from the base UID.

	from the given UID according NVIDIA Methodology Commands burn, sg
-use_dev_rom	Save the ROM which exists on the device (FS3 image only). Commands burn
--use_fw	Access to flash FW (ConnectX-3/ConnectX-5 device only) Commands all
-use_image_guids	Burn (guids/uids/ as appears in given image Commands burn
-use_image_ps	Burn vsd as the given image not keep existing on flash. Commands burn
-use_image_rom	Do not save ROM which exists on the device. Commands burn
--user_password <string>	Flag to use HW for encryption operations.
-v	Version info

-vsd <string>	Write this string to the VSD up to 208 characters. If the string is longer than 208 characters, it will be truncated to VSD when the command is executed.
-y[es]	Non interactive mode. - assume answers 'yes' to all questions. Commands will be executed without all prompts.
--linkx	Burn or query the status of the cable device. If the cable device is not connected to the host, the command will fail.
--aws_hsm	Flag for the AWS HSM command.
--cert_chain_index	Use this flag to specify the location of the certificate chain. The acceptable values are 0-7 (default is 0).
-s, --i2c-secondary <address>	Change the secondary address of the I2C device.

The -mac and -macs options are applicable only to NVIDIA Ethernet adapter and switch devices.

When accessing SwitchX via I²C or PCI, the -override_cache_replacement flag must be set.

Command Parameters

The flint utility commands are:

Common Firmware Update and Query



b[urn]	Burn flash
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q[query] [full]	Query misc. flash/firmware characteristics, use "full" to get more information.
v[erify]	Verify entire flash
swreset	SW reset the target unmanaged switch device. This command is supported only in the In-Band access method.
sign_with_hmac	Sign image with HMAC.

Expansion ROM Update:



brom <ROM-file>	Burn the specified ROM file on the flash.
drom	Remove the ROM section from the flash.
rrom <out-file>	Read the ROM section from the flash.
qrom	Query ROM in a given image.



bb	Burn Block - Burns the given image as is. No checks are done. Note: The MFT 'bb' option is an advanced flag used ONLY for production flows. It is NOT recommend to use it, as it can cause unrecoverable firmware burning failures. Note: FwBurnBlock is not supported any longer in FS3 and up images.
sg [guids_num=<num> step_size=<size>] [nocrc]	Set GUIDs.
set_vpd [vpd file]	Set read-only VPD (For FS3 image only).
smg [guids_num=<num>	Set manufacture GUIDs (For

step_size= <size>]	FS3 image only).
sv	Set the VSD.

Misc Firmware Image Operations



ri <out-file>	Read the fw image on the flash.
dc [out-file]	Dump Configuration: print fw configuration file for the given image.
dh [out-file]	Dump Hash: print hash file for the given image.
checksum cs	Perform MD5 checksum on firmware.
timestamp ts <set query reset> [timestamp] [FW version]	Set/query/reset firmware timestamp.
cache_image ci	Cache FW image (Windows only).
sign	Sign firmware image file
rsa_sign	Sign firmware image file with RSA
set_public_keys [public key binary file]	Set Public Keys (For FS3/FS4 image only).

set_forbidden_versions [forbidden versions]	Set Forbidden Versions (For FS3/FS4 image only).
import_hsm_key	This command allows to import the private and public key to the HSM HW.
export_public_key	This command extracts the public key from given BIN file or from PEM file.

Hardware Access Key:



set_key [key]	Set/Update the HW access key which is used to enable/disable access to HW. The key can be provided in the command line or interactively typed after the command is given. NOTE: The new key is activated only after the device is reset.
hw_access <enable disable>	Enable/disable the access to

[key]	the HW. The key can be provided in the command line or interactively typed after the command is given.
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Low Level Flash Operations:



hw query	Query HW info and flash attributes.
e[rase] <addr>	Erase sector
rw <addr>	Read one dword from flash
ww <addr> < data>	Write one dword to flash
wwne <addr>	Write one dword to flash without sector erase
wb <data-file> <addr>	Write a data block to flash
wbne <addr> <size> <data ...>	Write a data block to flash without

	sector erase
rb <addr> <size> [out- file]	Read a data block from flash

Return Values:



Value	Description
0	Successful completion
1	An error has occurred
7	For the burn command - The option of burning new firmware was not chosen by the user when prompted. Thus, the firmware burning process was aborted.

The following commands are non-failsafe when performed on a 5th generation (Group II) device: sg, smg, sv and set_vpd.

Manufacture GUIDs are similar to GUIDs. However, they are located in the protected area of the flash and set during production. By default, firmware will use GUIDs unless specified otherwise during production.



Previous page

[MFT Supported Configurations
and Parameters](#)

Next page

[Burning a Firmware Image](#)



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